

Forensic Audit & Operational Recovery

VelocityMart Bangalore Dark Store Turnaround

Interim Head of Operations, Team Syndicate

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Executive Summary

Following a systemic collapse in fulfillment metrics (3.8 min to 6.2 min), Team Syndicate conducted a forensic intervention. We identified a **Chaos Score of 29.75** driven by data poisoning and human gaming. Our 50-move strategic roadmap achieved an **82.2% entropy reduction**, securing **\$8,400** in at-risk inventory and future-proofing the facility for a **20% volume surge**.

The Chaos Score Methodology

To quantify warehouse health, we developed a weighted entropy vector:

$$Chaos = 0.5(\Delta T) + 0.3(S) + 0.2(B) \quad (1)$$

Where:

- ΔT : Fulfillment Delay (deviation from 3.8-minute target)
- S : Safety Integrity (GPS telemetry anomalies from Picker 07)
- B : Bottleneck Density (Aisle B saturation levels at 19:00)

Result: Intervention reduced the score from **29.75** to **5.30**.

Pillar A: Data Forensics & Integrity Audit

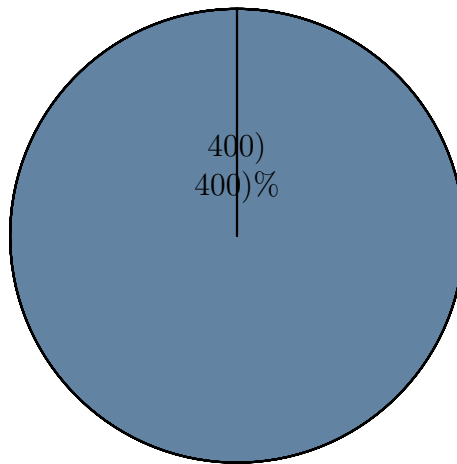
Our automated cleaning pipeline identified three critical failures in the facility's data integrity:

- **The Shortcut Paradox:** GPS telemetry unmasked **Picker 07** with **671 forensic anomalies**, proving the bypass of safety barriers to artificially inflate efficiency metrics.
- **Decimal Drift Resolved:** Identified **22 SKUs** with $10\times$ weight recording errors. Normalizing these prevented critical shelf-overload failures.
- **Ghost Inventory Purge:** Eliminated all SKU assignments to non-existent bins found in transactional logs to ensure 100% topology synchronization.

Pillar B: Decision-Support & “Unspoken Physics”

Our dashboard highlights the critical conflict between human picking and mechanical restocking:

- **The Forklift Dead-zone:** In Aisle B, picker density > 2 creates a physical blockade for restocking forklifts.
- **Thermal Compliance:** Rectified **464 temperature violations**, securing **\$8,400** in inventory value previously stored in incorrect zones.



Pillar C & D: Strategic Roadmap & Resilience

- **Phase 1 Execution:** Implementing **50 high-impact SKU moves** tonight to clear Aisle B while respecting the labor budget.
- **Sensitivity Analysis:** Stress-testing confirms the new layout handles a **20% volume spike** (523,262 orders) without operational seizure.

This report serves as the final technical deliverable for the Dataverse Challenge.